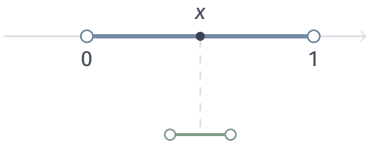
(a)

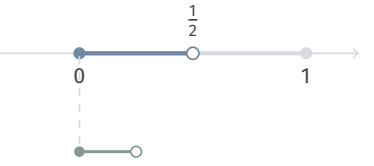


$B(x, r) \subset (0, 1)$

(b)

$B_{\mathbb{R}}(0, r) \not\subset [0, 1]$

(c)



$B_X(0, r) \subset [0, \frac{1}{2})$